EXERCISES 4.3

Answers to selected odd-numbered problems begin on page ANS-13.

4.3.1 Complex Trigonometric Functions

In Problems 1–8, express the value of the given trigonometric function in the form $a + ib$.

1. $\sin(4i)$

2. $\cos(-3i)$

3. $\cos(2 - 4i)$

4. $\sin\left(\frac{\pi}{4} + i\right)$

5. $\tan(2i)$

6. $\cot(\pi + 2i)$

7. $\sec\left(\frac{\pi}{2} - i\right)$

8. $\csc(1 + i)$

In Problems 9–12, find all complex values z satisfying the given equation.

9. $\sin z = i$

10. $\cos z = 4$

11. $\sin z = \cos z$

12. $\cos z = i \sin z$

In Problems 13–16, verify the given trigonometric identity.

13. $\sin(-z) = -\sin z$

14. $\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$

15. $\overline{\cos z} = \cos \bar{z}$

16. $\sin\left(z - \frac{\pi}{2}\right) = -\cos z$

In Problems 17–20, find the derivative of the given function.

17. $\sin(z^2)$

18. $\cos(ie^z)$

19. $z \tan \frac{1}{z}$

20. $\sec\left(z^2 + (1 - i)z + i\right)$

4.3.2 Complex Hyperbolic Functions

In Problems 21–24, express the value of the given hyperbolic function in the form $a + ib$.

21. $\cosh(\pi i)$

22. $\sinh\left(\frac{\pi}{2} i\right)$

23. $\cosh\left(1 + \frac{\pi}{6} i\right)$

24. $\tanh(2 + 3i)$

In Problems 25–28, find all complex values z satisfying the given equation.

25. $\cosh z = i$

26. $\sinh z = -1$

27. $\sinh z = \cosh z$

28. $\sinh z = e^z$

In Problems 29–32, verify the given hyperbolic identity.

29. $\cosh^2 z - \sinh^2 z = 1$

30. $\sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2$

31. $|\sinh z|^2 = \sinh^2 x + \sin^2 y$

32. $\operatorname{Im}(\cosh z) = \sinh x \sin y$

In Problems 33–36, find the derivative of the given function.

33. $\sin z \sinh z$

34. $\tanh z$

35. $\tanh(iz - 2)$

36. $\cosh\left(iz + e^{iz}\right)$

Focus on Concepts

37. Recall that Euler's formula states that $e^{i\theta} = \cos\theta + i\sin\theta$ for any real number θ. Prove that, in fact, $e^{iz} = \cos z + i\sin z$ for any complex number z.

38. Solve the equation $\sin z = \cosh 2$ by equating real and imaginary parts.

39. If $\sin z = a$ with $-1 \le a \le 1$, then what can you say about z? Justify your answer.

40. If $|\sin z| \le 1$, then what can you say about z? Justify your answer.

41. Show that all the zeros of $\cos z$ are $z = (2n+1)\pi/2$ for $n = 0, \pm 1, \pm 2, \ldots$.

42. Find all z such that $|\tan z| = 1$.

43. Find the real and imaginary parts of the function $\sin \bar{z}$ and use them to show that this function is nowhere analytic.

44. Without calculating the partial derivatives, explain why $\sin x \cosh y$ and $\cos x \sinh y$ are harmonic functions in $\mathbf{C}$.

45. Prove that $\sin z$ is a one-to-one function on the domain $-\pi/2 < x < \pi/2, -\infty < y < \infty$.

46. Use the identity $\cos z = \sin\left(z + \frac{1}{2}\pi\right)$ to find the image of the region $-\pi \le x \le 0$ under the mapping $w = \cos z$. Describe the images of vertical and horizontal lines in the region.

47. Use the identity $\sinh z = -i\sin(iz)$ to find the image of the region $-\pi/2 \le y \le \pi/2$, $-\infty < x < \infty$, under the mapping $w = \sinh z$. Describe the images of vertical and horizontal lines in the region. [*Hint*: The identity implies that $w = \sinh z$ is a composition of linear mappings and the complex mapping $w = \sin z$.]

48. Find the image of the region defined by $-\pi/2 \le x \le \pi/2$, $y \ge 0$, under the mapping $w = (\sin z)^{1/4}$, where $z^{1/4}$ represents the principal fourth root function.

49. Find the period of each of the following complex functions.

(**a**) $\cosh z$ (**b**) $\sinh z$ (**c**) $\tanh z$

50. Find all zeros of each of the following functions.

(**a**) $\cosh z$ (**b**) $\sinh z$

51. Find the real and imaginary parts of the functions $\cosh z$ and $\sinh z$.

52. Use the results of Problem 51 to find formulas for $|\cosh z|$ and $|\sinh z|$.

53. Verify the following identities.

(**a**) $\sin(z+\pi) = -\sin z$ (**b**) $\cos(z+\pi) = -\cos z$

54. Use the identities in Problem 53 to show that $\tan z$ is a periodic function with a real period of π.

4.4 Inverse Trigonometric and Hyperbolic Functions

To The Instructor: In this section, we introduce the inverse complex trigonometric and hyperbolic functions. This material can be skipped without affecting the development of topics in subsequent chapters.

INTRODUCTION The complex logarithmic function $\ln z$ was defined in Section 4.1 to solve equations of the form $e^w = z$. Because the complex exponential function is periodic, there are infinitely many solutions to such equations, and, consequently, $\ln z$ is necessarily a multiple-valued function. In this section we repeat this process for equations involving the complex trigonometric and hyperbolic functions. Because the complex trigonometric and hyperbolic functions are periodic, their inverse functions are multiple-valued. Furthermore, since the complex trigonometric and hyperbolic